

Technology Stack

Date	15 March 2026
Team ID	XXXXXXXX
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap 5	Used to build a responsive, interactive, and user-friendly web interface for collecting user information and displaying well-being prediction results, charts, and recommendations.

2	Backend / Server-Side	Python 3.x, Flask	Flask provides a lightweight framework for handling user requests, integrating the machine learning model, processing data, and generating prediction results efficiently.
3	Database / Data Storage	CSV Dataset, SQLite (or MySQL if you are using MySQL)	Stores user information, training dataset, and prediction history. SQLite is lightweight and easy to integrate with Flask applications.
4	Cloud / Hosting / Deployment	Render / PythonAnywhere / Local Flask Server	Used to deploy the Flask web application so users can access the well-being prediction system through a web browser.
5	Version Control & CI/CD	Git, GitHub	Enables version control, collaboration among team members, code backup, and project management throughout development.

6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib	Used for data preprocessing, Linear Regression model training, visualization of results, and saving/loading the trained machine learning model.
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